Contact

melidisc@gmail.com

www.linkedin.com/in/christos-melidis (LinkedIn)

Top Skills

Prototyping

Engineering

Communication

Languages

English (Professional Working)

German (Limited Working)

Greek (Native or Bilingual)

Publications

KURE: Kinematic universal remote interface a human centred remote robot control paradigm

KURE: a Two-Way Adaptive System for Intuitive Robot Control

Intuitive control of mobile robots: an architecture for autonomous adaptive dynamic behaviour integration

A Human Centric Approach to Robotic Control

An Exploration on Intuitive Interfaces for Robot Control Based on Self Organisation

Patents

Systems and methods for generating automatic training suggestions

Systems and methods for generating a chatbot

Christos Melidis

Staff ML Scientist @ Ada | Generative AI, Conversational AI, NLP

Greece

Summary

I'm a Machine Learning Scientist with over a decade of experience in AI, specializing in Conversational AI, including Speech, Natural Language Processing and Generation. With a PhD in Computer Science, focused on innovative Neural Network architectures for Robotics, I excel at transforming cutting-edge research into practical, real-world applications.

My role often bridges research and production, guiding projects from idea to deployment. This involves everything from initial concept development and requirements gathering with Product teams, to hands-on coding with Engineering teams. I have extensive experience in developing various Neural Architectures like RNNs, LSTMs, DNNs, CNNs, and more, often building these systems from the ground up to achieve outstanding results.

Driven by a passion for innovation and a commitment to excellence, I strive to stay at the leading edge of Generative and Conversational AI, delivering solutions that make a difference.

Experience

Ada

4 years 6 months

Staff Machine Learning Scientist

February 2024 - Present (2 years)

- AI Agent Platform: Architected a modular, LLM-based system that scales effortlessly, maximizes performance, and reduces costs by allocating optimal models per module.
- Context Management Redesign: Drove end-to-end roadmap for knowledge-base retrieval—lifted top-5 article match from 80%→90%, yielding a 3% absolute increase in "Resolved Inquiries" across 5M monthly conversations and slashing hallucinations by 2%.

- Playbooks Framework: Led ML effort to auto-generate Standard Operating Procedures, establishing the "adherence rate" metric and shipping with <1% error in adherence—driving further gains in resolution rates.
- Leading efforts on Testing Methodologies & Performance Evaluation.
- Leading efforts and projects on AI Explainability methods.

Senior Machine Learning Scientist

August 2021 - February 2024 (2 years 7 months)

Member of Ada's ML Research team, working on the forefront of Conversational and Generative AI.

AI Automation & Optimization

• ATS (Automatic Training Suggestions): AI-driven training automation (patented).

• Autolaunch: Developed a declarative bot creation framework (patented).

Generative AI

• Generative Replies: Enhanced AI response generation with retrieval, moderation, and query rewriting.

• Generative Actions: Optimized tool usage and procedural automation.

Linked Business

Machine Learning Consultant

April 2020 - April 2022 (2 years 1 month)

Thessaloniki, Central Macedonia, Greece

Linked Business is a FinTech company collecting company data from official state sources and providing B2B sales leads and KYC services to its customers

Serve as a trusted advisor on artificial intelligence, machine-learning, and text-mining strategies, translating the latest academic research and industry trends into actionable insights for stakeholders.
- Design and prototype innovative algorithms, then collaborate cross-functionally to embed them seamlessly into the company’s products and platforms.
- De-risk new ideas by leading rapid proof-of-concept initiatives that validate technical feasibility and business value.
- Distill complex challenges into clear product requirements, and shape data-driven feature roadmaps that accelerate time-to-market and ROI.

Outcome: Empowered teams to transform cutting-edge AI research into scalable, high-impact product capabilities.

Omilia - Conversational Intelligence
Senior Machine Learning Engineer - R&D
February 2020 - August 2021 (1 year 7 months)
Remote

Member of Omilia’s R&D team, having applied research on Anti-Spoofing, Emotion Recognition, Diarization and other classification tasks based on voice data.
Tech: Python, PyTorch, C++, PyTorch Lightning, Transformer models, Attention models

Curation Zone
Data Scientist
April 2019 - February 2020 (11 months)
Remote

Working on AI technologies to identify the World’s best filmmakers. Deep Neural
Networks, Natural Language Processing, Scoring and Data Management.
Tech: Python, BERT, ELMO, LSTMs, TensorFlow, RESTful API, MongoDB

Stella Novus Limited
Lead AI Developer
October 2018 - March 2019 (6 months)

Working on a digital marketing platform [http://www.ainu.ai/], using Deep Neural
Networks, Natural Language Processing.
Tech: Python, LSTMs, TensorFlow, RESTful API

i-Dat
AI & Machine Learning Scientist
September 2017 - September 2018 (1 year 1 month)
Remote

Working on a Chatbot engine with text comprehension, using Neural Networks, Natural Language Processing and Natural Language Generation.
Tech: Python, LSTMs, Keras, DialogFlow, Rasa

University of Naples Federico II
Research Scientist
June 2017 - September 2017 (4 months)

Researcher Position under the supervision of prof. Davide Marocco at NAC (Natural and Artificial Cognition) Lab. Working on a Generative Adversarial Architecture based on Echo State Networks, with applications to music generation.
Tech: Python, LSTMs, Echo State Networks,Theano

Plymouth University
3 years 4 months

Researcher PHD Student
April 2014 - July 2017 (3 years 4 months)

PhD Fellow in Plymouth, research fellow at CogNovo Doctoral Programme, funded by the EU Marie Curie initiative and Plymouth University. (http://www.cognovo.eu/)

Research Project Title: ADAPTIVE NEURAL ARCHITECTURES FOR INTUITIVE ROBOT CONTROL

Taking inspiration from human robot interaction, ergonomic principles, and autonomous robotics this project proposes a human-centric framework for robot control inspired by the current advancements in machine learning, artificial intelligence and autonomous robotics. The interface is realised as a software agent connecting the two end points of the system: human and robot, providing an adaptive and intelligent interface for robot control.
In our studies we have been able to demonstrate the proposed methodology for both self-exploration of robotic morphologies and acquisition of human behaviours.
On the methodological part, we work with Artificial Neural Architectures, and specially with Recurrent Neural Networks for the connection of the system with the user. The adaptive mechanisms implemented on the robot are also modeled as Neural Networks, as is the dynamical system guiding the robot’s autonomy. In particular we experiment with Recurrent Neural Networks with Parametric Bias (RNNPB), Echo State Networks and Reservoir Networks, Feed-Forward Neural Networks, Deep Neural Networks, as well as Convolutional Neural Networks.

Laboratory Assistant
2015 - 2017 (2 years)
Plymouth, United Kingdom

Cross-platform application development in C++ Lab sessions (SOFT336SL). Final year Computer Science BSc level students. Under the supervision of Associate Professor Davide Marocco.

Research Assistant
March 2016 - April 2016 (2 months)
Plymouth, United Kingdom

Under the guidance of Professor Michael Hyland and Professor Susan Denham, responsible for
the delivery of research support to the project: Machine Learning in Functional Disorders:
exploratory study.

Laboratory Assistant
2015 - 2016 (1 year)
Plymouth, United Kingdom

Teaching Parallel and Distributed Programming Lab sessions (SOFT339). Final year Computer Science BSc level students. Under the supervision of Associate Professor Davide Marocco.

Laboratory Assistant
2015 - 2016 (1 year)
Plymouth, United Kingdom

Teaching the Robotics and Control Lab sessions (MECH533). Final year Mechanical Engineering MSc level students. Under the supervision of Professor Tony Belpaeme.

Research Assistant
October 2015 - December 2015 (3 months)
Plymouth, United Kingdom

Under the guidance of Professor Mike Phillips. This research has been used to develop a new
prototype open source sentiment analysis tool for arts and culture discourse, providing a
practical test of the initial findings about automated social media analysis. A Machine Learning
Algorithm for Natural Language Processing was developed, capable of real time operation and
understanding of spoken language as observed through Tweets.

Prof. I.A. TSOUKALAS, MEP
Information System Officer

November 2010 - April 2014 (3 years 6 months)

Education

University of Plymouth

Doctor of Philosophy (PhD), Artificial Intelligence & Robotics · (2014 - 2017)

Aristotle University of Thessaloniki (AUTH)

Bachelor of Science (BSc), Computer Science · (2008 - 2013)